PIC to SAP S/4HANA: document types and postings

This note covers three PIC messages that now have a live SAP mapping: contracts (OUTLINE_AGREEMENT), the reversal of a returned payment (PAYMENT_REVERSAL) and supply-chain finance (SCF_FINANCING).

All values below are defaults. Each one is a tenant rule. A rule can be set per company code and has an effective date. Change it in Administration, Rules (no code change).

Everything here was tested against a mock SAP only. The client must confirm it in their SAP sandbox before go-live. Section 6 lists what to confirm.

Code: `backend/app/pic/platform/sap_mapping.py` (request bodies), `platform/sap_defaults.py` (rules and checks), `platform/connectors.py` (class `SapS4`), `platform/sap_customizing.py` (the simulator's checks). Mock SAP data: `backend/data/erp_mock/SAP_CONFIG/`.

1. Mapping table

PIC document	SAP document	SAP document type	SAP API (entity set)	Number SAP returns
Purchase requisition	Purchase requisition	NB	API_PURCHASEREQ_PROCESS_SRV (A_PurchaseRequisitionHeader)	PurchaseRequisition
Purchase order	Purchase order	NB (FO, UB by setting)	API_PURCHASEORDER_PROCESS_SRV (A_PurchaseOrder)	PurchaseOrder
Quantity contract	Quantity contract	MK	API_PURCHASECONTRACT_PROCESS_SRV (A_PurchaseContract)	PurchaseContract
Value contract	Value contract	WK	API_PURCHASECONTRACT_PROCESS_SRV (A_PurchaseContract)	PurchaseContract
Rate card	Value contract	WK	API_PURCHASECONTRACT_PROCESS_SRV (A_PurchaseContract)	PurchaseContract
Statement of work (SOW)	Value contract	WK	API_PURCHASECONTRACT_PROCESS_SRV (A_PurchaseContract)	PurchaseContract
Scheduling agreement	Scheduling agreement	LP	API_SCHED_AGRMT_PROCESS_SRV (A_SchAgrmtHeader)	SchedulingAgreement
Master agreement	Not sent	-	-	-
NDA	Not sent	-	-	-
Payment returned by the bank	Reversal of the clearing document	reason 01	Client's payment service (A_VendorPaymentReversal)	ReverseDocument
Supply-chain finance election	Journal entry, supplier to financier	KA	Client's journal service (A_JournalEntry) or IDoc FIDCCP02	AccountingDocument

Rate card and SOW: both are sent as WK value contracts. The contract's target value is the header target value (PurchaseContractTargetAmount). Each PIC line is an item. A line without a material is a text item with a material group and account assignment category U (unknown), so the call-off order assigns the cost centre. If the client uses service items, set `serviceItemCategory` (for example D for lean services or 9 for services with service lines).

Master agreements and NDAs are legal documents. SAP has no object for them. PIC never sends them. This is not an error: nothing is queued and no retry happens.

2. What happens over a contract's life

PIC event	What PIC sends	SAP call
Contract activated	Create	POST of the header with its items (deep insert).
Amendment, extension or index reset (new version or new prices)	Change	PATCH of each existing item, POST of new items, then PATCH of the header.
Termination takes effect	Close	PATCH of the header: validity end = last day in force.
Renewal by a successor that starts early	Close of the predecessor	PATCH of the header: validity end = the day before the successor starts.
Renewal by a successor	Create of the successor	A new SAP agreement.
Contract expires	Nothing	The SAP validity end already ends it.
Master agreement or NDA	Nothing	PIC never queues it. A call-off type a rule lists in notSent is recorded as ERP_NOT_SENT in the audit trail.

PIC writes its reference into the header field CorrespncExternalReference ("Your reference", 12 characters). A change and a close write the new reference in the last call. So if a call fails half way, PIC does not find the change in SAP and sends it again. Item PATCHes are safe to repeat. A new item that already reached SAP is PATCHed on the retry, not created twice.

Versions reach SAP in order. A newer version waits while an older one is still retrying.

A contract migrated from SAP keeps its SAP number. Its first change from PIC updates that number.

3. Payment reversal

When the bank returns a payment, PIC asks SAP to reverse the payment's clearing document.

Fields sent: CompanyCode, AccountingDocument (the clearing document), FiscalYear, ReversalReason, PostingDate, ResetClearing, DocumentHeaderText (PIC reference), PaymentReference, ReversalText.

- Reason 01 (reversal in current period) with the payment's original posting date. This is the default.
- If PIC has closed that posting period, reason 02 (reversal in closed period) with the return date.
- ResetClearing true: the service first resets the cleared items (FBRA), then reverses (FB08).
- Fiscal year: the year the payment's fiscal year starts in (April start: January 2027 is fiscal year 2026).

SAP has no released OData API that resets clearing and reverses in one step. So the reversal goes to the client's own payment service, the one named in `PIC_ERP_SAP_S4_PAYMENT_SERVICE` (the same service that posts outgoing payments). Without that service the reversal is not mapped. PIC then tells treasury to reverse it in SAP by hand. IDoc transports do not map it either.

4. Supply-chain finance (SCF)

SAP has no standard API for this. PIC posts one journal entry:

Line	Posting key	Account	Debit / credit	Amount
1	21	Supplier (vendor)	Debit	Invoice gross
2	31	Financier business partner (vendor), due on the maturity date	Credit	Invoice gross
2 (alternative)	50	Financier G/L account	Credit	Invoice gross

Document type KA. Reference: invoice number in DocumentReferenceID, the SAP invoice number in AssignmentReference, PIC reference in DocumentHeaderText.

Transport: the client's journal entry service named in `PIC_ERP_SAP_S4_JOURNAL_SERVICE` (fields follow the Journal Entry Post API). Without it, PIC sends IDoc FIDCCP02.

This posting needs the client's design. Open points:

- Clearing: the journal entry does not clear the supplier invoice. The client decides how the supplier's invoice and the debit line are cleared (automatic clearing by assignment, or a clearing posting).
- Fees and discount charged by the financier are not posted.
- Withholding tax (TDS) is not split. The full gross is moved.
- Whether the financier is a vendor (business partner) or a G/L liability account.

5. Rules and defaults

Rules are effective-dated and can be scoped per company code. A version must hold every key it needs: the most specific version wins on its own and is not merged with the tenant-wide one. Keys left out take the code default.

SAP-OUTLINE-AGREEMENT

Key	Default	Meaning
documentTypes	QUANTITY_CONTRACT MK, VALUE_CONTRACT WK, SCHEDULING_AGREEMENT LP, RATE_CARD WK, SOW WK	PIC type to SAP document type.
contractTypes	MK, WK	Types sent to the purchase contract API (category K).
schedulingAgreementTypes	LP, LPA	Types sent to the scheduling agreement API (category L).
notSent	MASTER_AGREEMENT, NDA	Never sent.
purchasingOrg	empty	Empty: `PIC_ERP_SAP_S4_PURCH_ORG`, else the only active PIC purchasing org of the company code.
purchasingGroup	empty	Empty: `PIC_ERP_SAP_S4_PURCH_GROUP`, else the only PIC purchasing group of that org.
textItemMaterialGroup	empty	Material group of lines without a material. Required in live mode when such lines exist.
textItemAccountAssignment	U	Account assignment category of text items.
serviceItemCategory	empty	Item category of text items of the serviceItems types (D, 9 ...).
serviceItems	RATE_CARD, SOW	Types whose text items get serviceItemCategory.
closeOnTermination	true	Send the validity end when a contract is terminated.
closeOnRenewal	true	Send the validity end when a successor starts before the end.

SAP-PAYMENT-REVERSAL

Key	Default	Meaning
reversalReason	01	SAP reversal reason (table T041C).
postingDate	ORIGINAL	ORIGINAL (payment posting date) or RETURN_DATE (day the return was recorded). RETURN_DATE needs a reason that allows another date.

Key	Default	Meaning
closedPeriodReason	02	Used, with the return date, when PIC has closed the payment's period. Empty = never.
resetClearing	true	Reset the cleared items before the reversal.

SAP-SCF-FINANCING

Key	Default	Meaning
documentType	KA	FI document type.
postingKeys	supplierDebit 21, financierCredit 31, glCredit 50	Posting keys.
financiers	empty	Per PIC financier code (rule EARLY-PAYMENT scf.financier, default PROGRAMME): `businessPartner` or `glAccount`, and `paymentMethod`.

In live mode an SCF election whose financier is not configured is rejected and retried, so the client sees it in the integration monitor. The simulator uses its mock financier instead.

Connector settings (environment)

Setting	Use
PIC_ERP_SAP_S4_PAYMENT_SERVICE	Client OData service for payments and their reversal.
PIC_ERP_SAP_S4_JOURNAL_SERVICE	Client OData service for the SCF journal entry.
PIC_ERP_SAP_S4_TERMS_MAP	PIC payment terms to SAP terms key for contracts, e.g. `N45=Z045`. Unmapped terms are not sent; SAP then uses the vendor master.
PIC_ERP_SAP_S4_COMPANY_MAP	PIC company code to SAP company code.

Mock values (dataset pack D21)

Pack `sap_config` (D21) writes these rules for AXN and KAC from `SAP_CONFIG/PIC_RULES.json`: purchasing org and group per company code (AXN 1000 PO10/G01, 1100 PO30/G04; KAC 1000 PO10/G03, 2000 PO21/G01), material group SERVICES for text items, reason 01 / 02, and the mock financier 9800001 (AXN) or 9800011 (KAC). They are MOCK values.

6. What the client must confirm in the sandbox

1. Document types: are MK, WK and LP used, or own types (for example ZMK)? Add own types to contractTypes or schedulingAgreementTypes and to documentTypes.
2. Rate card and SOW: value contract WK with text items, or service items (item category D or 9)?
3. Field names we could not confirm from public documentation:
   - contract item text `PurchaseContractItemText` and price `ContractNetPriceAmount`;
   - item category `PurchasingDocumentItemCategory` and `AccountAssignmentCategory` on contract items;
   - scheduling agreement header type `PurchasingDocumentType`, item text `PurchasingDocumentItemText`, price `NetPriceAmount`;
   - whether CorrespncExternalReference can be changed by PATCH on both APIs;
   - whether item keys are sent as `'10'` or `'00010'`.
4. Time-dependent prices: PIC sends the current price as the item net price. If the client keeps price validity periods as conditions (PB00 with validity), that needs the condition entity of the API.
5. Lines removed in PIC are not deleted in SAP (PIC does not remove contract lines today).
6. Termination: is setting the validity end enough, or should items also get the deletion indicator or a block?
7. Purchasing organisation and group per company code.
8. Payment terms map (PIC terms to SAP terms keys).
9. Reversal: the client's payment service must reset clearing and reverse. Confirm reason 01 (no other posting date allowed) and reason 02 in table T041C.
10. Payment documents over IDoc: the number PIC receives is an IDoc number, not the accounting document. A reversal needs the real clearing document number, so use the payment service.
11. SCF: the whole design in section 4 (clearing, fees, tax, vendor or G/L, posting keys, document type, the financier's business partner and its company code extension).
12. Services: the names `ZAPI_VENDOR_PAYMENT_SRV` and `ZAPI_JOURNAL_ENTRY_SRV` are placeholders. The entity sets and field names we use are in `SAP_CONFIG/samples/`.

7. Simulator

Without `PIC_ERP_SAP_S4_URL` the SAP connector is a simulator. For these three messages it runs the same mapping as the live connector. It then checks the request against the mock customizing: document type allowed (T161, T003), reversal reason and posting date (T041C), financier and its company code (BP_FINANCIER), payment method (T042Z), material group (T023), and the mandatory fields. It refuses what SAP would refuse, and the message goes to retry with the reason. A contract change or close keeps the agreement's number, as in SAP. The request sent is kept with the simulated document (`erp_document.payload.sapRequest`).